

- 6 A particle P moving in a straight line has displacement x m from a fixed point O on the line and velocity v m s⁻¹ at time t s. The acceleration of P , in m s⁻², is given by $6v\sqrt{v+9}$. When $t = 0$, $x = 2$ and $v = 72$.
- (a) Find an expression for v in terms of x . [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (b) Find an expression for x in terms of t . [5]

[illegible]